

## Devices that can be controlled

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|-------------------------|-----------------------------------------------------------------------|
| <b>IR devices</b>       | Televisions, air-conditioners, radios, music systems                  |
| <b>Digital devices</b>  | Computers, game consoles, mobile phones, tablets                      |
| <b>Energy controls</b>  | Thermostats, smartfans, vents, in-floor meetings, fireplaces          |
| <b>Security</b>         | Door locks, smartglasses, window shades, CCTV cameras, control panels |
| <b>Wiring</b>           | Outlets, switches, dimmers                                            |
| <b>Exterior devices</b> | Garage doors, sprinkler systems                                       |

## For more information

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|----------------|---------------------------------------------------------------------------------------|
| <b>Device</b>  | Prime                                                                                 |
| <b>Company</b> | Tomar Technologies Pvt Ltd                                                            |
| <b>Website</b> | <a href="http://tomartechnologies.com/Tomar1">http://tomartechnologies.com/Tomar1</a> |

multiple devices and functions using simple finger gestures. Ring by Logbar is a similar product that lets you control systems with a ring worn on a finger. All these devices are in the pre-order stage, some of these firms having failed to bring out their products almost a year after their promised delivery dates and online crowd funding.

However, the application field of similar devices available in the market is very fragmented. For example, Fitbit is basically for personal health monitoring and Ring is more into home automation. As a user, if I have to get five types of devices for five different functions, that leads to more gadgets, more cost and increased confusion.

## Why Prime

The idea behind Prime was to unify all these solutions into a single, small form factor wearable device. The basic set of sensors used in different types of gesture-recognition devices is the same. Team Tomar wanted to create a gadget that could control almost all electronic devices and could be useful for the customer from the time he or she woke up in the morning to when he or she went to bed at night, from increasing the speed of a fan while sitting on a couch to serving as an access card in the office and to continuously monitoring health parameters.

The problem Prime is trying to

solve is to make it natural for users to interact with devices. For example, if you are on a call and the television volume is too high, you can extend your hand with your palm facing the television (stop signal) in order to mute it. So, Prime identifies the device you are pointing at and identifies the action of the hand in order to perform the intended function.

Many automation devices these days let you control appliances using a customisable mobile app. But a mobile app can be complicated as you have to open the app first and then change the settings. Also, there are different apps for different applications; for example, one app to control the television and another to control the security of the home.

When you point at a device, the line-of-sight is identified in order to select the device. If there are two devices identified in the same line-of-sight, Prime identifies the nearest device. The team is also working on algorithms for identifying a single device from a stack of devices. Gesture recognition is one of the proprietary algorithms that Tomar Technologies has developed and applied patent for.

## Spreading across different sectors

As mentioned earlier, Prime would be finding applications across multiple sectors, namely, home automation, gaming, corporate world, and care for the elderly and physically challenged. There are several use cases. People can find their own use cases once they start interacting with this wearable wristband, the developers expect.

Home automation is one segment the team is looking at, as they see a lot of potential in this field in coming years. Tomar is currently planning to tie up with a company in security business. They are also trying to target the corporate sector, hospitality industry and healthcare. In the next two years, the team is planning to develop a healthcare package where vitals of the user will be continuously measured, pulled to the Cloud, analysed and a warning message sent to the doctor when something goes beyond normal.

Prime is currently in its prototype stage and extensive work is being done to reduce the form factor and increase the accuracy of the device. Developers of the product are expecting to get the first batch of Prime into the market by June 2015. The firm is looking at both B2B and B2C monetisation.

## Ongoing

Infra-red (IR)-enabled devices do not require any extra set of components for communication. But devices like light, fan and other home appliances have to be fitted with sensors and transmitters, which is done by automation firms. Tomar is partnering with system integrators, automation companies and security solutions providers in order to offer a complete automation solution.

In the B2C segment, the firm is developing a smartswitch (a combination of switch and IR receiver) for devices that are not IR or Bluetooth based. The consumer has to plug the switch in a normal socket and plug the appliance in the socket of this switch. The IR receiver is placed on the appliance, and when you plug it in using the smartswitch, you can control the device.

In short, the wearable band directly controls all IR and Bluetooth-enabled devices like laptops, gaming devices and air-conditioners, and indirectly, using the smartswitch, controls the devices that are not IR or Bluetooth enabled. ●